

Should traditional and digital financial literacy be reported together or separately? —Traditional and Digital Financial Literacy in Italy

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Abstract

This study uses data from the 2023 IACOFI Survey of Adult Financial Literacy released by the Bank of Italy to examine the relationship between traditional financial literacy and digital financial literacy, as well as the demographic characteristics and actual financial behaviour associated with mismatches between the two. Following the official scoring rules, the study first reconstructs separate indicators of traditional and digital financial literacy from the original questionnaire items. Survey-weighted means are then compared with the officially published results to verify the accuracy of the reconstruction. On this basis, we compare the overall relationship between the two forms of financial literacy and classify respondents into four financial-literacy profiles according to whether each indicator reaches a specified target. We then compare these groups by age, gender, education and region, and further examine differences in digital financial activity, online acquisition of financial products and adverse financial experiences. The results show a moderate positive correlation between traditional and digital financial literacy, but the two do not fully overlap: a substantial share of respondents reaches a high level in only one dimension. Differences by education are particularly pronounced, while neither digital financial participation nor adverse financial experiences have a simple one-to-one relationship with financial literacy. The findings suggest that, when analysing modern financial capability, reporting traditional and digital financial literacy separately is more useful than a single composite indicator for identifying differences across population groups and understanding their practical significance.

Financial literacy; Digital financial literacy; Financial literacy profiles; Digital finance; Italy

1 About the Dataset

This study uses data from the 2023 Survey of Adult Financial Literacy and Financial Competence in Italy (IACOFI 2023), published by the Bank of Italy (Banca d'Italia) [1]. The survey is based on the OECD/INFE financial-literacy framework [3] and covers financial knowledge, financial behaviour, financial attitudes, digital financial literacy, use of digital financial services and demographic characteristics. The microdata used in the project contain 4,862 respondents and 219 variables and include survey weights that support population-representative descriptive analysis. In accordance with the official methodology, analyses involving digital financial literacy mainly use the subsample of respondents with internet access [4].

2 Declaration

We declare that we used generative AI in many aspects of this work, mainly for discussing the project structure, debugging code, editing and formatting text, translation and language polishing.

3 Research question

3.1 Statement of the research question and core concepts

As financial services are increasingly delivered through the internet, mobile applications and digital platforms, financial knowledge and behaviour in their traditional sense can no longer fully capture the capabilities individuals need in a digital financial environment. This study therefore analyses traditional financial literacy and digital financial literacy as two related but not entirely equivalent concepts.

In this study, traditional financial literacy follows the official scoring framework used by the survey and consists of three components: financial knowledge, financial behaviour and financial attitudes. Financial knowledge primarily reflects an individual's understanding of basic financial concepts such as interest, inflation, risk and return, and risk diversification. Financial behaviour concerns practical conduct such as budgeting, saving, borrowing, selecting financial products and managing everyday finances. Financial attitudes reflect an individual's orientation towards consumption, saving and long-term financial planning. Together, these three components form the total traditional financial-literacy score. Digital financial literacy likewise consists of knowledge, behaviour and attitude dimensions, but its focus shifts to the digital financial environment. It covers, for example, knowledge of digital contracts, the commercial use of personal data and crypto-assets, as well as digital-security behaviours and attitudes such as protecting passwords, verifying financial-service providers and safeguarding personal financial information [3]. Digital financial literacy therefore means more than simply knowing how to use the internet or online banking: it reflects the ability to understand information, identify risks and behave appropriately in a digital financial environment.

On the basis of these two concepts, the core research question is:

To what extent do traditional and digital financial literacy overlap among Italian adults?
When the two forms of literacy do not coincide, what patterns does this difference display in demographic characteristics, digital financial participation and adverse financial experiences?

This central question can be divided into several connected analytical questions. First, how strong is the overall association between the two forms of financial literacy? Second, is there a substantial number of respondents who perform well in only one of the two dimensions? Third, are the resulting financial-literacy profiles distributed differently across age, education, gender and region? Finally, are these profiles associated with actual use of digital financial services and with adverse experiences such as fraud, unauthorised card use and unrecognised transactions?

To examine mismatches between the two forms of financial literacy more directly, respondents are further divided into four groups according to whether each indicator reaches a 70% target [4]: both targets met, only the traditional financial-literacy target met, only the digital financial-literacy target met, and neither target met. This classification moves the analysis beyond average correlations and makes it possible to identify mismatch groups that may be hidden by overall averages or by a single composite indicator.

3.2 Research motivation and potential stakeholders

The main motivation for studying this question is that, as financial activity becomes increasingly digital, a high level of traditional financial literacy does not necessarily mean that an individual also possesses sufficient knowledge, risk awareness and safe behaviour in a digital financial environment. Compressing all financial capability into a single indicator may conceal different kinds of capability gaps. For example, a person may understand interest, inflation and investment risk but lack the ability to recognise digital fraud, protect personal data or verify an online financial-service provider. Conversely, some people may be familiar with digital tools and cybersecurity rules while having relatively weak traditional financial knowledge.

Distinguishing between traditional and digital financial literacy therefore has practical significance. For central banks, financial regulators and public-policy makers, this type of analysis can show whether existing financial-education policies cover only traditional financial knowledge while overlooking new risks in the digital financial environment. For financial-education organisations, schools and consumer-protection groups, literacy mismatches across population groups can help in designing more targeted

educational content rather than assuming that all people with low financial literacy have the same capability deficit.

For banks, financial-technology firms and other providers of digital financial services, understanding the relationship between financial literacy and actual use of digital services may also help to improve product design, risk warnings and customer education. If people who use digital financial services do not necessarily have high digital financial literacy, simply expanding access to digital services will not by itself ensure their safe and effective use.

Finally, the issue is also relevant to researchers. Whether traditional and digital financial literacy are two manifestations of a common capability or two partly independent dimensions is an important question in the measurement of financial literacy. This study does not attempt to provide a final causal explanation with the available data. Instead, it first uses descriptive evidence to examine the degree of overlap, the structure of the differences and their relationship with real-world behaviour, thereby providing a basis for more detailed measurement research and causal analysis.

4 The Data

4.1 Data description

This study uses microdata from the 2023 Survey of Adult Financial Literacy and Financial Competence in Italy (IACOFI 2023), released by the Bank of Italy [1]. The Bank of Italy has conducted IACOFI every three years since 2017, using a methodological framework developed by the OECD International Network on Financial Education (INFE) [3]. Questions on digital financial competence were included for the first time in the 2023 survey, making this wave especially suitable for studying the relationship between traditional and digital financial literacy [1].

The target population of the 2023 survey comprised adults aged 18 to 79 living in Italy. The data were collected in February and March 2023 through computer-assisted telephone interviewing (CATI). The official survey uses post-stratification weights to make the sample more closely resemble the Italian adult population. The adjustment factors include geographical area, municipality size, gender, age and education [1].

This project uses the officially released Stata microdata file. The dataset contains 4,862 respondents and 219 variables, including the survey-weight variable `wght`. Each row corresponds to one respondent, while each column represents a questionnaire item, a demographic variable or another survey-related variable. The original numerical codes were retained when importing the data so that the financial-literacy indicators could be reconstructed according to the official OECD/Bank of Italy scoring rules [2, 3].

The dataset covers several main categories of information. The first consists of demographic characteristics, including gender, age, education and geographical region. The second consists of items related to traditional financial literacy, including financial knowledge, financial behaviour and financial attitudes. The third comprises digital financial-literacy items, including digital financial knowledge, digital-security behaviour and digital financial attitudes. The questionnaire also contains information on digital financial activities, financial products obtained entirely online, and adverse financial experiences such as fraud, data disclosure, unauthorised use of bank cards and unrecognised transactions. The official variable documentation defines each question and its coding in detail [2].

It is important to note that the official statistics on digital financial literacy cover only respondents who can access the internet [4]. Following this approach, the project defines respondents with `qd14` = 1 as having internet access [2], yielding 4,427 individuals. The main analyses comparing traditional and digital financial literacy therefore use this subsample rather than automatically assigning a digital financial-literacy score of zero to people without internet access.

In addition, the data used in the project do not contain ready-made total scores for Traditional Financial Literacy and Digital Financial Literacy. They contain responses at the questionnaire-item level [2]. Before conducting the substantive analysis, the two literacy indicators must therefore be reconstructed separately from the original items according to the official rules [3], and the reconstructed results must be checked against the published official statistics [1]. This step provides the foundation for all subsequent analyses.

4.2 Advantages and limitations of the data for the research objective

The dataset is well matched to the research question. First, the same survey contains all the items required to measure both traditional and digital financial literacy. The two capabilities can therefore be compared directly for the same respondents without combining data from different surveys. Second, the data include not only measurement items on financial knowledge and attitudes but also demographic characteristics, digital financial activities and adverse financial experiences. This allows the study to move beyond correlations between indicators and examine population characteristics and real financial behaviour. Third, the survey provides official weights, enabling weighted descriptions of means, proportions and group distributions that more closely reflect the structure of the target population than unweighted sample counts.

The data also have important limitations. First, IACOFI is a cross-sectional survey. The study can therefore describe associations between variables but cannot establish causal relationships. Even if people with higher financial literacy engage in more digital financial activities, for example, this does not justify the conclusion that financial literacy causes greater digital participation. Income, occupation, age, education and the actual need for financial services may affect both.

Second, many variables are self-reported and may be affected by recall error, differences in interpretation and social-desirability bias. In particular, reported financial behaviour, digital activities and adverse experiences are not necessarily equivalent to directly observed behavioural records.

Third, the main analytical sample for digital financial literacy includes only internet users [4]. Conclusions about the relationship between traditional and digital literacy therefore apply strictly to people with internet access and cannot automatically be generalised to all adults. A complete lack of internet access may itself represent an important form of digital exclusion, but it is treated here as a separate issue rather than being coded directly as low digital financial literacy.

Fourth, analyses of fraud, unauthorised card use and other adverse financial experiences require particular caution. The probability of encountering a problem may be related not only to financial literacy but also to an individual's exposure to digital financial services. Even a person with high digital financial literacy who frequently uses online banking, digital payments and online financial products may be more likely to report at least one adverse experience simply because of greater exposure. These results are therefore more appropriately treated as descriptive comparisons and should not be interpreted as evidence of a direct protective effect of financial literacy.

Finally, the study uses the financial-literacy items and scoring framework defined by the OECD and the Bank of Italy [1, 3]. Their official source and standardised methodology improve reproducibility and comparability, but they do not automatically demonstrate that the items fully capture the complex concept of financial literacy. The project therefore uses the official scoring framework as its operational foundation and subsequently asks whether the indicators distinguish groups with different demographic characteristics and actual behaviours; it does not treat the official indicators as the sole or absolute measure of real financial capability.

5 Methodology

5.1 Overall analytical logic

The study adopts a progressively structured descriptive framework. Rather than beginning by calculating a correlation between traditional and digital financial literacy, it first reconstructs the two sets of indicators from the original questionnaire items according to the OECD/INFE scoring rules used by the Bank of Italy [2, 3]. It then validates the reconstruction by comparing the resulting statistics with the official results [1]. Only after confirming that the scoring procedure reproduces the published results does the study investigate the relationship between the two forms of financial literacy. The full analytical process can be summarised as follows:

Raw questionnaire data → Literacy-score reconstruction → Validation →
 Traditional–Digital comparison → Literacy profiles → Demographic analysis →
 Digital activity and adverse experiences

The purpose of this order is to avoid conducting statistical analysis before the meaning and implementation of the indicators have been verified, and to ensure that every step builds on the preceding one. First, comparable Traditional Financial Literacy and Digital Financial Literacy indicators are

established. Second, their overall association is examined. Third, that overall relationship is decomposed into different combinations at the individual level. Finally, the distribution of these combinations across demographic groups and actual digital financial activity is analysed.

The study relies primarily on descriptive and associational analysis. Its objective is not to identify causal effects, but to determine the extent to which traditional and digital financial literacy overlap and whether mismatches between them correspond to meaningful differences in population characteristics and behaviour.

5.2 Reconstruction of the traditional financial-literacy indicator

The dataset does not directly provide the complete individual-level Traditional Financial Literacy score required for this study. The indicator is therefore reconstructed from the original questionnaire responses according to the official scoring rules [2, 3]. Traditional financial literacy consists of three dimensions:

$$\text{Traditional Financial Literacy} = \text{Knowledge} + \text{Behaviour} + \text{Attitude}.$$

Financial knowledge ranges from 0 to 7, financial behaviour from 0 to 9, and financial attitude from 0 to 4. The total traditional financial-literacy score therefore satisfies:

$$0 \leq \text{Traditional Score} \leq 20.$$

Financial knowledge. The financial-knowledge component consists of seven scored items. They cover basic concepts including inflation and purchasing power, interest on loans, simple interest, compound interest, the relationship between risk and return, the definition of inflation and risk diversification. In principle, each correct answer receives one point, while an incorrect answer, “do not know” response or refusal receives zero. Under the official rule, the compound-interest item is evaluated jointly with the preceding simple-interest question and earns a point only if the previous answer is also correct. The seven items are summed to produce a financial-knowledge score from 0 to 7.

Financial behaviour. The financial-behaviour component has a maximum score of 9 and covers budgeting, active saving, borrowing when income is insufficient, selection of financial products and day-to-day financial management. The budgeting item requires the respondent to be responsible for household or personal money and to use at least two budgeting practices. Active saving is scored according to whether the respondent used at least one saving method during the previous 12 months. The borrowing item considers whether the respondent dealt with insufficient income by relying on loans, credit-card overdrafts, unpaid bills or similar means. Selection of financial products can contribute up to two points. A higher score is awarded when an independent source of information is used; a lower score is awarded when products are compared or other information sources are consulted. Four further behavioural items assess whether the respondent keeps track of financial affairs, sets long-term financial goals, considers affordability before making a purchase and pays bills on time. Summing all items gives a financial-behaviour score from 0 to 9.

Financial attitude. Financial attitude is calculated from three Likert items scored from 1 to 5. In accordance with the official rule, invalid responses are replaced with the midpoint value of 3. The mean of the three items is then calculated and reduced by one, giving an attitude score from 0 to 4. Because the calculation uses an average rather than a count of correct answers, both the attitude score and the resulting total traditional financial-literacy score may contain decimals. The knowledge, behaviour and attitude components are finally added to obtain each respondent’s Traditional Financial Literacy score.

5.3 Reconstruction of the digital financial-literacy indicator

Digital Financial Literacy is likewise divided into three components under the official framework [3]:

$$\text{Digital Financial Literacy} = \text{Knowledge} + \text{Behaviour} + \text{Attitude}.$$

Digital financial knowledge ranges from 0 to 3, digital financial behaviour from 0 to 4 and digital financial attitude from 0 to 3. Thus:

$$0 \leq \text{Digital Score} \leq 10.$$

Digital financial knowledge is measured through three items concerning whether digital contracts must have a handwritten signature, whether publicly shared personal data may be used for commercial profiling, and whether crypto-assets constitute legal tender. Each correct answer receives one point.

Digital financial behaviour consists of four items that mainly assess digital financial-security practices: avoiding the sharing of passwords or PINs with others, checking whether financial-service providers are regulated, avoiding the public disclosure of personal financial information and changing passwords regularly.

The digital financial-attitude component also contains three items. These address the risk of conducting transactions over public Wi-Fi, the importance placed on website security, and attitudes towards the terms and conditions of digital transactions. Responses indicating safer or more cautious attitudes are scored as one point, taking the direction of each item into account.

The three components are summed to produce a Digital Financial Literacy score from 0 to 10.

An important methodological treatment concerns internet access. The Bank of Italy’s digital financial-literacy statistics cover respondents with internet access [1, 4], so this study uses **qd14** to identify that group [2], comprising 4,427 respondents. Only this common analytical sample is used in the main Traditional–Digital comparisons. Respondents without internet access are not simply assigned a digital financial-literacy score of zero, because “having no internet access” and “having internet access but low digital financial literacy” are distinct concepts.

5.4 Validation of the reconstructed indicators

Before conducting the substantive analysis, the study first tests whether the reconstructed scores reproduce the official results published by the Bank of Italy [1]. Because IACOFI is a sample survey, weighted means are calculated using the survey weight **wght**:

$$\bar{x}_w = \frac{\sum_{i=1}^n w_i x_i}{\sum_{i=1}^n w_i},$$

where x_i is the individual’s score and w_i is the survey weight assigned to that respondent.

The knowledge, behaviour, attitude and total scores for traditional financial literacy are calculated for all respondents. In accordance with the official methodology, the four digital financial-literacy statistics are calculated only for respondents with internet access. The eight weighted means are then compared with the corresponding means published by the Bank of Italy.

The purpose of this validation is not to demonstrate that the official scoring framework is the only or optimal method for measuring financial literacy. Rather, it confirms that the study has correctly interpreted and implemented the official scoring rules. Only after the published results can be reproduced reliably do the subsequent comparisons of traditional and digital financial literacy rest on a consistent measurement basis.

5.5 Initial comparison of traditional and digital financial literacy

Traditional Financial Literacy has an original range of 0–20, whereas Digital Financial Literacy ranges from 0–10. To facilitate direct interpretation and visualisation, both indicators are first converted linearly to a 0–100 scale:

$$\text{Traditional \%} = \frac{\text{Traditional Score}}{20} \times 100, \quad \text{Digital \%} = \frac{\text{Digital Score}}{10} \times 100.$$

This transformation changes only the scale. It does not alter respondents’ rankings and does not imply that the two indicators have been combined into a composite measure.

To measure the overall association between the two indicators, the study uses Spearman’s rank correlation coefficient. Spearman correlation is preferred to reliance on Pearson’s linear correlation because the main question is whether the two financial-literacy scores exhibit an overall monotonic relationship, without assuming in advance that the relationship is strictly linear.

The Spearman coefficient may be expressed as the consistency between the two individual rankings:

$$\rho_s = \text{Corr}(R_T, R_D),$$

where R_T and R_D are the ranks of the traditional and digital financial-literacy scores, respectively. A scatter plot is also used to display the joint distribution of the two indicators at the individual level, so that the entire relationship is not reduced to a single coefficient.

5.6 Four financial-literacy profiles

A correlation coefficient describes whether two indicators tend to move together overall, but it does not show what kinds of mismatch occur at the individual level. Following the overall association analysis, the study therefore constructs financial-literacy profiles. Using the adopted 70% target [4], each respondent is classified according to whether the minimum target for traditional and digital financial literacy is met. This yields four mutually exclusive and exhaustive combinations:

Traditional	Digital	Profile
$\geq 70\%$	$\geq 70\%$	Both meet target
$\geq 70\%$	$< 70\%$	Traditional only
$< 70\%$	$\geq 70\%$	Digital only
$< 70\%$	$< 70\%$	Neither meets target

Table 1

The purpose of this procedure is not to replace the original continuous scores with four profiles. It supplements the continuous-indicator analysis by identifying, in particular, the two mixed profiles: Traditional only and Digital only. If only the average correlation were reported, or if a single unified score were constructed, people who score highly in one dimension and poorly in the other could be concealed. Profile analysis is therefore used to examine the structure of mismatches between the two forms of financial literacy.

The overall proportion of each profile is calculated using survey weights:

$$\text{Share}_g = \frac{\sum_{i \in g} w_i}{\sum_i w_i} \times 100.$$

This produces group proportions adjusted by survey weighting rather than simple unweighted sample frequencies.

5.7 Demographic comparisons

After constructing the four profiles, the study examines whether financial literacy is distributed evenly across population groups. The comparison variables are gender, age, education and geographical region. Some original survey variables are regrouped to make the resulting categories sufficiently interpretable. Age is divided into four groups: 18–34, 35–49, 50–64 and 65–79. For respondents who reported an age band rather than an exact age, the midpoint of the corresponding interval is used as an approximation. The ten original education categories in the questionnaire are consolidated into three broader categories [2]:

Tertiary, Upper secondary, Lower secondary or less.

The five macro-geographical regions are further consolidated into:

North, Centre, South and Islands.

For each demographic group, the study calculates survey-weighted mean Traditional and Digital scores to observe overall differences across groups.

Means, however, may conceal different combinations of traditional and digital literacy within a group. Following the comparison of average scores, the weighted proportions of the four literacy profiles are therefore also calculated within age and education groups. This makes it possible to observe both between-group differences in average scores and the within-group composition of literacy, avoiding an interpretation based solely on means.

5.8 Comparison of digital financial activity

After the demographic analysis, the study examines whether different literacy profiles correspond to different levels of actual digital financial participation.

Two behavioural indicators that are separate from the literacy scores are used. The first is the Digital Activity Count. Seven types of digital financial activity are selected from the questionnaire [2]; an activity receives a value of one if it was undertaken sometimes, often or very often during the previous 12 months. The values are then summed to form:

$$\text{Digital Activity Count} \in [0, 7].$$

This indicator reflects the breadth of types of digital financial activity in which respondents participate, rather than their exact frequency of use.

The second indicator is the Online Product Count. The questionnaire includes five categories of financial products that may have been obtained entirely online [2]. One point is awarded for each category, so that:

$$\text{Online Product Count} \in [0, 5].$$

For each of the four literacy profiles, the survey-weighted mean of both indicators is calculated.

The purpose of this analysis is not to incorporate digital financial use into the literacy score. Instead, use is treated as behavioural information outside the indicators in order to examine whether different literacy profiles correspond to different degrees of actual digital financial participation.

5.9 Comparison of adverse financial experiences

Finally, the study investigates whether observable differences exist between literacy profiles in their experience of digital financial risk.

The questionnaire covers four types of adverse experience during the previous two years: fraud; phishing or data disclosure; unauthorised bank-card use; and unrecognised transactions [2].

To create a simple and interpretable indicator, a respondent is assigned:

$$\text{Any Adverse Experience} = 1$$

if at least one of the four experiences is reported. If none is reported, then:

$$\text{Any Adverse Experience} = 0.$$

The survey-weighted proportion reporting at least one adverse experience is then calculated for each literacy profile. This comparison is descriptive only. Adverse experiences may be related not only to financial literacy but also to the degree of use of digital financial services. Greater use may increase potential exposure to fraud, unauthorised card use and unrecognised transactions. Differences between profiles and adverse experiences are therefore not interpreted as evidence that financial literacy has a causal protective effect; only their joint distribution in the data is described.

5.10 Analytical tools and the reasons for their selection

All data processing and statistical analysis were completed in a Python 3.11 environment. The full analytical workflow was retained in a Jupyter Notebook to make the steps from reading the raw data to calculating indicators and producing final figures and tables reproducible.

`pandas` was used to read the Stata data, recode variables, perform grouped calculations and generate result tables. `NumPy` was used for numerical calculations and conditional scoring logic. The `spearmanr` function from `SciPy` was used to calculate Spearman's rank correlation between traditional and digital financial literacy. `Matplotlib` and `Seaborn` were used to create scatter plots, ordinary bar charts and stacked bar charts. `pathlib` was used to manage paths for raw data, result tables and image outputs consistently.

These tools were selected because the dataset is moderate in size and the core tasks are transparent and reproducible questionnaire recoding, weighted descriptive statistics and visualisation rather than the training of complex predictive models. The analysis therefore prioritised interpretability,

reproducibility and the relationship between each step and the research question. Machine-learning models that were not needed to answer the question were not introduced merely to increase technical complexity.

All key intermediate results were saved as separate CSV tables, while the main figures were saved as high-resolution PNG files. The Notebook therefore serves both as the analytical environment and as a complete reproducibility record, allowing the scoring rules, sample selection, variable transformations and statistical calculations to be inspected and rerun step by step.

6 Results

6.1 Validation of reconstructed indicators against official results

Before comparing traditional and digital financial literacy, the study tests whether indicators reconstructed from the original questionnaire items reproduce the statistics published by the Bank of Italy. The reconstructed survey-weighted means are very close to the official results [1]. The mean scores for traditional financial knowledge, behaviour and attitude are 3.694, 4.676 and 2.308, respectively, compared with official values of 3.7, 4.7 and 2.3. The total traditional financial-literacy score is 10.679, compared with the official result of 10.7. The survey-weighted means for digital financial knowledge, behaviour and attitude are 1.280, 2.086 and 1.236, compared with the official values of 1.3, 2.1 and 1.2. The total digital financial-literacy score is 4.602, compared with the official result of 4.6.

Indicator	Weighted mean in this study	Official mean
Traditional knowledge (0–7)	3.694	3.700
Traditional behaviour (0–9)	4.676	4.700
Traditional attitude (0–4)	2.308	2.300
Traditional total (0–20)	10.679	10.700
Digital knowledge (0–3)	1.280	1.300
Digital behaviour (0–4)	2.086	2.100
Digital attitude (0–3)	1.236	1.200
Digital total (0–10)	4.602	4.600

Table 2

All differences are small, and the official results themselves are reported to only one decimal place. The comparison therefore indicates a high degree of consistency between the reconstruction in this study and the official calculations. The Traditional and Digital literacy scores used in the subsequent analysis can consequently be considered operationally consistent with the official definitions. This result validates the accurate reproduction of the scoring rules; it does not demonstrate that the scoring system itself is the only or optimal way to measure financial literacy.

6.2 Overall relationship between traditional and digital financial literacy

Within the common analytical sample of 4,427 respondents with internet access [1, 4], both indicators are converted to a 0–100 scale and Spearman’s rank correlation coefficient is calculated. The result is:

$$\rho_s = 0.443.$$

The corresponding p-value is:

$$p = 4.83 \times 10^{-212}.$$

The sample therefore displays a clear positive relationship: respondents with higher traditional financial literacy also tend to have higher digital financial literacy. The correlation is nevertheless well below one, indicating that the two are not simple repetitions of the same dimension.

The extremely small p-value is not the most informative result. With a large analytical sample, even a moderate association can easily reach conventional statistical significance. The coefficient itself

is more meaningful: a value of 0.443 indicates moderate co-movement but still leaves substantial differences at the individual level.

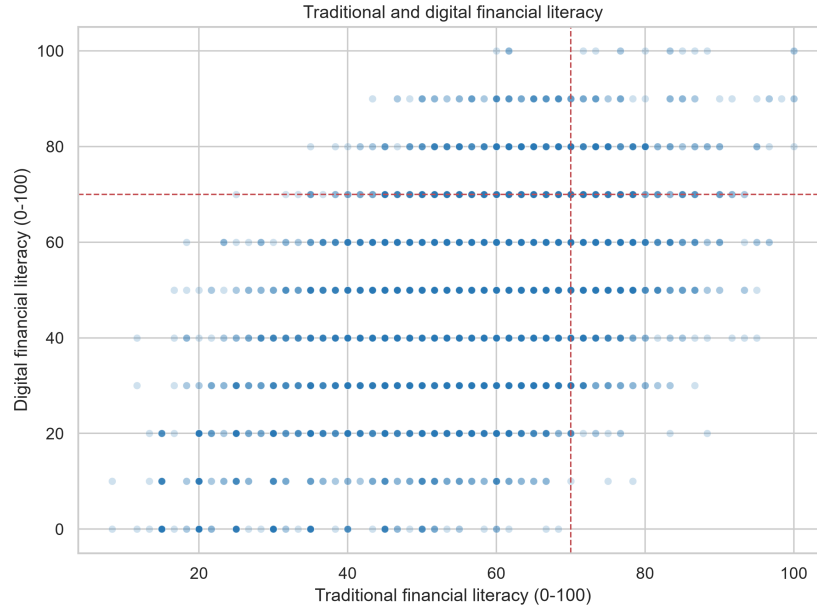


Figure 1: Traditional and digital financial literacy

The scatter plot further illustrates this point. An overall upward-right trend is visible, but respondents are not concentrated along a narrow line. Similar Traditional scores are associated with substantially different Digital scores, and vice versa.

Simply answering whether the two indicators are correlated is therefore insufficient to describe their relationship. The next step is to investigate how common such mismatches are at the individual level.

6.3 Four financial-literacy profiles

Respondents are divided into four mutually exclusive literacy profiles according to whether each indicator reaches the 70% target [4]. The survey-weighted results are:

Literacy profile	Weighted proportion
Both meet target	5.84%
Traditional only	11.70%
Digital only	14.37%
Neither meets target	68.09%

Table 3

The largest group is Neither meets target, accounting for approximately 68.09%. Only 5.84% of respondents meet both the Traditional and Digital literacy targets. More important for the research question are the two mixed profiles: 11.70% meet only the traditional financial-literacy target, while 14.37% meet only the digital target. Together they account for:

$$11.70\% + 14.37\% = 26.07\%.$$

In other words, approximately one quarter of the analytical population displays a clear Traditional–Digital mismatch.

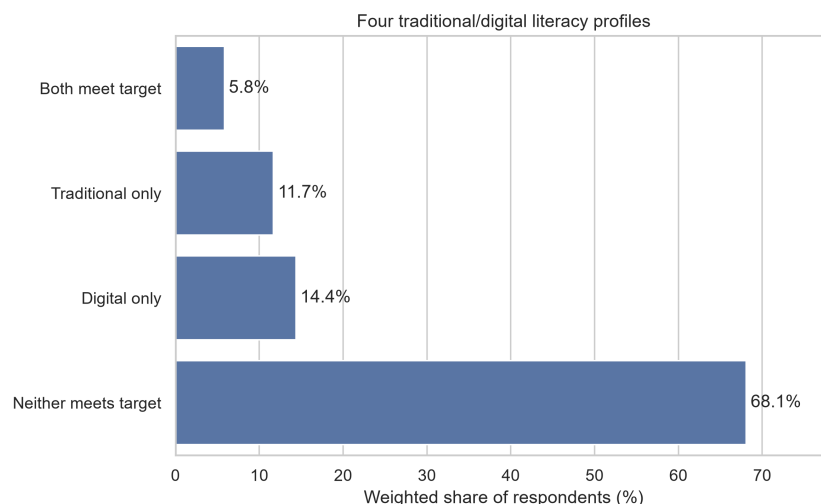


Figure 2: Four financial-literacy profiles

This finding gives the correlation reported in the previous section a more intuitive interpretation. Traditional and Digital literacy are positively correlated overall, but they cannot substitute for one another. If a single unified financial-literacy score were constructed, the Traditional-only and Digital-only groups, which may have different needs, could be combined and concealed.

6.4 Differences in financial literacy across demographic groups

The next analysis compares weighted mean Traditional and Digital literacy scores across gender, age, education and regional groups. The results are as follows:

Characteristic	Group	Traditional	Digital
Gender	Men	55.51	46.95
	Women	53.49	45.08
Age	18–34	50.04	42.61
	35–49	56.69	47.64
	50–64	55.84	46.45
	65–79	55.01	47.69
Education	Tertiary	59.54	51.42
	Upper secondary	54.45	45.40
	Lower secondary or less	48.03	40.64
Region	North	55.96	47.56
	Centre	54.92	45.84
	South and Islands	52.26	44.02

Table 4: Traditional and digital financial literacy by demographic group

First, the Traditional score exceeds the Digital score in every demographic group. Lower digital financial literacy is therefore not concentrated in one particular population segment; it is present across the sample. Gender differences are relatively small. Men’s Traditional and Digital scores are 55.51 and 46.95, respectively, while women’s are 53.49 and 45.08, a difference of roughly two percentage points for each indicator.

Age does not display a simple pattern in which younger people have higher digital financial literacy. The 18–34 group actually has the lowest average score on both indicators, with 50.04 for Traditional and 42.61 for Digital. The means for the 35–49, 50–64 and 65–79 groups are relatively close. The

Digital score of 47.69 among respondents aged 65–79 is slightly higher than that of the other age groups.

The clearest differences are found across education levels. The Traditional and Digital scores of respondents with tertiary education are 59.54 and 51.42, while the corresponding scores among those with lower secondary education or less are only 48.03 and 40.64. Both indicators follow the pattern:

Tertiary > Upper secondary > Lower secondary or less.

The Traditional score differs by approximately 11.5 percentage points between the highest and lowest education groups, while the Digital score differs by approximately 10.8 points.

There are also regional differences: both mean scores are highest in the North, followed by the Centre, and lowest in the South and Islands. These gaps are nevertheless smaller than those associated with education. The Notebook’s summary of this stage likewise identifies education as the most pronounced demographic difference and regional differences as comparatively modest.

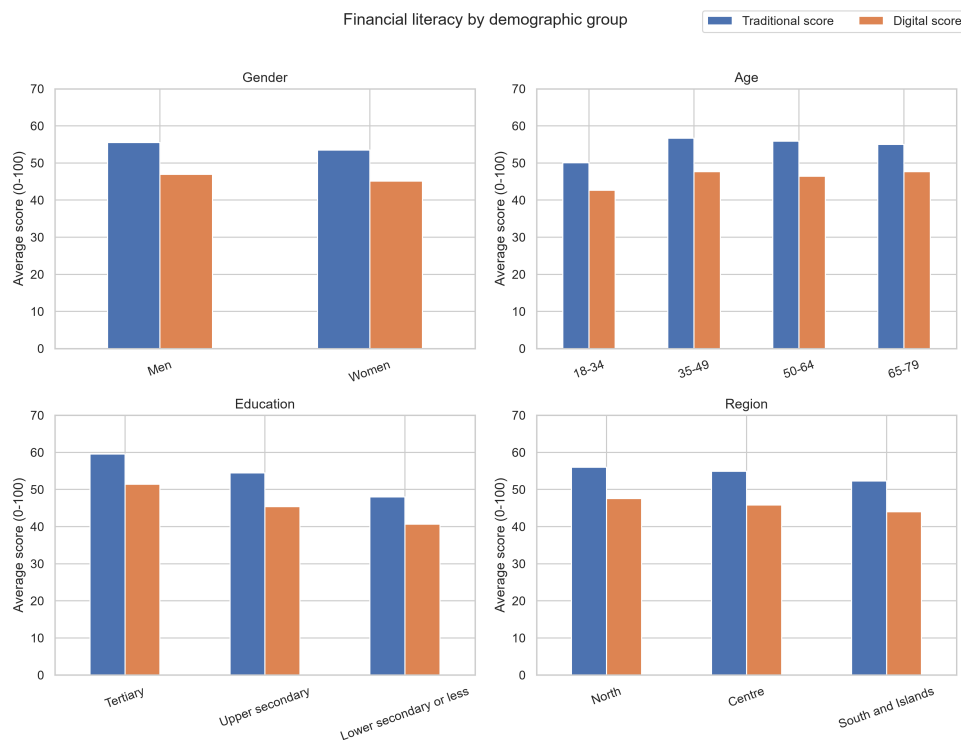


Figure 3: Financial literacy by demographic group

These comparisons still describe only group averages and cannot show whether Traditional and Digital literacy combine in the same way within each group. The composition of the four profiles across demographic groups must therefore be examined.

6.5 Literacy profiles within age and education groups

The profile composition of age groups shows that the two mixed profiles, Traditional only and Digital only, occur in every age group.

Literacy profile	Weighted proportion
Both meet target	5.84%
Traditional only	11.70%
Digital only	14.37%
Neither meets target	68.09%

Table 5

Among respondents aged 18–34, 72.50% are in the Neither group, the highest proportion among the four age categories, while only 3.34% meet both targets. The 35–49 group has the lowest Neither proportion, at 64.52%, and the highest Both proportion, at 6.83%. Notably, a substantial Digital-only profile is present in every age group; no age category is entirely dominated by a single literacy profile. The relationship between age and the Traditional–Digital mismatch is therefore not simply linear.

Profile differences are clearer across education groups. Among respondents with tertiary education, 9.92% meet both targets and 56.51% meet neither. As education declines, the Both proportion falls steadily while the Neither proportion rises. Among respondents with lower secondary education or less, 76.96% meet neither target. Even within this lower-education group, however, an important mismatch remains:

Digital only = 15.46%,

whereas:

Traditional only = 4.74%.

A lower level of education therefore does not imply that every dimension of financial capability declines in the same proportion. Even in a group with lower overall average scores, some respondents reach the Digital target without reaching the Traditional target.

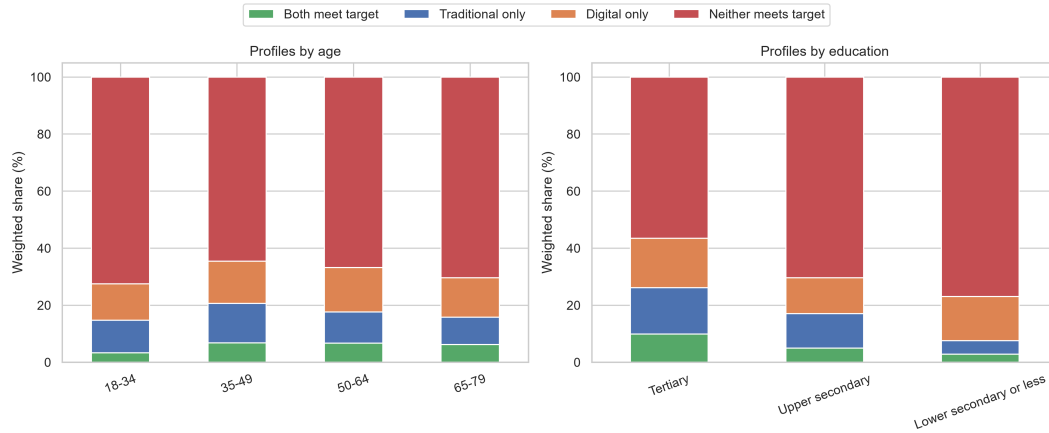


Figure 4: Financial profiles by age and education

This result further supports reporting Traditional and Digital literacy separately rather than replacing them with a single total score. Both mixed profiles are present in every age and education group.

6.6 Literacy profiles and digital financial activity

After analysing demographic characteristics, we examine whether literacy profiles are associated with actual digital financial participation. The survey-weighted means for the four groups are:

Profile	Digital activities (0–7)	Online products (0–5)
Both meet target	4.23	0.86
Traditional only	4.15	0.62
Digital only	4.14	0.66
Neither meets target	3.90	0.45

Table 6

The Both group has the highest level of digital financial participation, using an average of 4.23 types of digital financial activity and having obtained 0.86 types of financial product entirely online. The Neither group has the lowest means, at 3.90 and 0.45, respectively.

This suggests an association between literacy profile and actual digital financial participation. The difference between the Both and Neither groups is especially visible in financial products obtained online. The two mixed profiles, however, have almost identical numbers of digital activities:

$$\text{Traditional only} = 4.15, \quad \text{Digital only} = 4.14.$$

The extent to which digital financial services are actually used therefore cannot be explained simply by either literacy score alone. The Notebook offers the same interpretation: the nearly identical activity counts of the two mixed profiles suggest that service use is not determined by only one form of financial literacy. Actual digital financial participation may also be influenced by income, occupation, age, needs, product availability and personal technology habits. Because the study does not estimate a causal model, these differences are treated as descriptive results only.

6.7 Literacy profiles and adverse financial experiences

Finally, we compare whether members of the four literacy profiles experienced at least one instance of fraud, data disclosure, unauthorised bank-card use or an unrecognised transaction during the previous two years. The results are:

Profile	At least one adverse experience
Both meet target	13.34%
Traditional only	15.48%
Digital only	16.13%
Neither meets target	15.24%

Table 7

The proportions across the four groups range from 13.34% to 16.13%. The Both group has the lowest proportion, at 13.34%, but the Neither group is not the highest, at 15.24%. The highest proportion is found in the Digital-only group, at 16.13%. The data therefore do not support a simple pattern of:

$$\text{Higher literacy} \Rightarrow \text{Lower adverse experience.}$$

This result must be interpreted with particular care. Population groups do not have equal opportunities for exposure to digital financial risk. Individuals who use digital financial services more frequently may have both higher financial literacy and greater risk exposure. A higher adverse-experience rate therefore does not automatically imply that the group is more careless or has lower financial literacy.

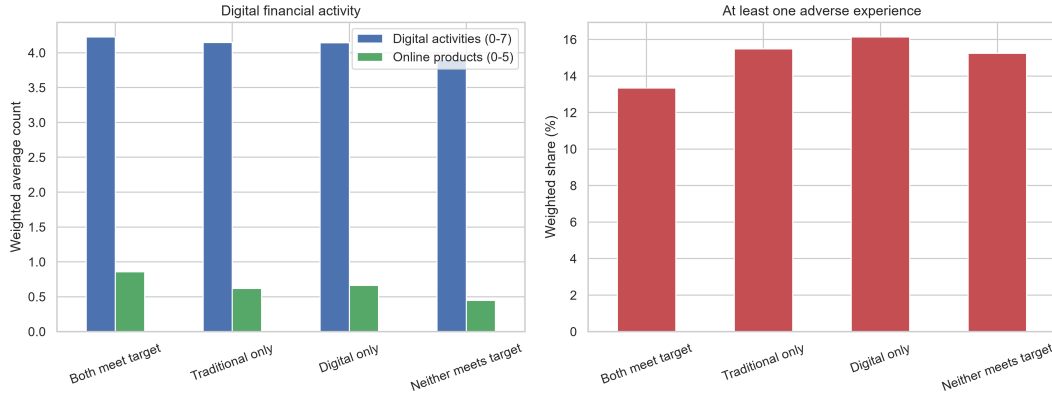


Figure 5: Digital financial activity and adverse experiences by literacy profile

6.8 Integrated interpretation of the main results

Overall, the central result of this study is not that Traditional Financial Literacy and Digital Financial Literacy are independent, but that they are related without being interchangeable. First, the two indicators have a moderate positive correlation:

$$\rho_s = 0.443,$$

showing that traditional and digital financial capability share a common foundation. Nevertheless, approximately 26% of respondents belong to the Traditional-only or Digital-only mixed profiles, demonstrating that this overall association still conceals substantial mismatch at the individual level.

Second, demographic characteristics are associated with both forms of financial literacy, and education displays the clearest differences. Even within the same age or education category, however, both mixed profiles are present. Demographic characteristics can therefore explain part of the overall variation, but they cannot fully explain why Traditional and Digital literacy diverge for particular individuals.

Third, literacy profiles are associated to some extent with digital financial activity. Respondents meeting both targets display the highest overall digital participation, while those meeting neither display the lowest. Yet actual activity is almost identical between the Traditional-only and Digital-only groups. Digital financial use therefore cannot be predicted simply from one literacy dimension.

Finally, adverse financial experiences do not follow a simple pattern in which higher literacy corresponds to fewer risk-related problems. This may reflect the complexity of the relationship between financial literacy and risk outcomes, as well as differences in exposure to digital financial services.

Taken together, the results suggest that Traditional and Digital financial literacy are best understood as related dimensions that should nevertheless be observed separately. Retaining both not only reveals differences in overall levels but also identifies Traditional-only and Digital-only groups that would easily be hidden in a single composite indicator. The Notebook’s descriptive summary similarly emphasises that mixed profiles are common across demographic groups; reporting the two forms of literacy separately therefore preserves more meaningful information than replacing them with one indicator.

7 Conclusion

This study uses the 2023 Bank of Italy IACOFI survey to examine the relationship between traditional and digital financial literacy among Italian adults and to analyse the demographic characteristics, digital financial participation and adverse financial experiences associated with mismatches between them [1]. Following the official OECD/Bank of Italy scoring rules, the study first reconstructs the Traditional Financial Literacy and Digital Financial Literacy indicators from the original questionnaire items [2, 3]. The survey-weighted means of the reconstructed components are very close to the official published results, showing that the official scoring system has been reproduced accurately and providing a consistent measurement basis for the subsequent analysis.

On this basis, the study finds a clear but incomplete positive relationship between traditional and digital financial literacy. Spearman’s rank correlation coefficient is 0.443, indicating that respondents with higher traditional financial literacy generally also have higher digital financial literacy, but that the two are not simple repetitions of a single capability. Their incomplete overlap becomes clearer in the literacy-profile analysis. Approximately 11.70% of respondents meet only the traditional financial-literacy target and 14.37% meet only the digital target; together, these two mixed profiles account for approximately 26.07%. Roughly one quarter of internet users therefore exhibit a pronounced Traditional–Digital mismatch.

Demographic comparisons further show that this mismatch is not distributed uniformly. Education presents the clearest gradient: respondents with tertiary education have higher mean scores on both Traditional and Digital literacy, while respondents with less education score substantially lower. Regional differences also exist, although they are smaller than differences by education. At the same time, the Traditional-only and Digital-only mixed profiles appear across age and education groups. Even people with similar demographic backgrounds do not necessarily display matching levels of the two forms of financial literacy.

When literacy profiles are linked to actual digital financial activity, respondents meeting both targets show the highest overall level of digital participation, while those meeting neither target show the lowest. The number of digital financial activities is nevertheless almost identical in the Traditional-only and Digital-only groups. Actual use of digital financial services therefore cannot be explained simply by either literacy score alone.

For adverse experiences such as fraud, unauthorised bank-card use and unrecognised transactions, the four profiles do not exhibit a simple monotonic relationship. The proportion reporting at least one adverse experience ranges from 13.34% to 16.13%, and higher financial literacy does not correspond consistently to a lower risk rate. Because the degree of digital financial use also affects opportunities for risk exposure, this finding should be treated as a descriptive association rather than evidence of a direct protective effect of financial literacy.

7.1 Gaps between the research objective and the results

The central objective of the study is not merely to determine whether Traditional Financial Literacy and Digital Financial Literacy are correlated. It is also to ask whether they can substitute for one another at the individual level and whether mismatches between them have practical meaning. From this perspective, the available data answer the first part of the research question relatively well. The two indicators have a moderate positive correlation, but a substantial share of respondents reaches the target in only one dimension. Combining them entirely into one unified indicator would therefore result in some loss of information.

For the second part of the research question, the analysis provides preliminary but limited evidence. Demographic analysis shows real differences in average levels and profile composition across groups. Analysis of digital financial activity also reveals some differences in actual participation across literacy profiles. The available data and methods, however, cannot explain why these differences arise. The study cannot determine whether education itself causes higher financial literacy, nor whether higher literacy directly encourages the use of digital financial services. Likewise, the complex relationship between adverse financial experiences and literacy may be influenced by digital financial exposure, income, occupation and other uncontrolled factors. The study is consequently better suited to answering:

“What structures and associations do these variables display?”

rather than:

“Does one form of financial literacy cause a particular behaviour or outcome?”

7.2 Strengths and limitations of the study

One major strength of the study is the clear progression of the analytical process. Rather than comparing unvalidated composite indicators, the study reconstructs both literacy scores from the original questionnaire items and first validates the calculations against official statistics. This improves the transparency and reproducibility of the entire workflow.

A second strength is that the relationship between Traditional and Digital literacy is not described using only a correlation coefficient. The four literacy profiles decompose the overall association at the individual level and explicitly identify the Traditional-only and Digital-only groups that averages can conceal. The Notebook’s descriptive summary likewise notes that mixed profiles are widespread across demographic groups and that reporting the two literacy dimensions separately preserves more information.

A third strength is that the analysis does not remain internal to the indicators. It also introduces demographic characteristics, digital financial activities and adverse financial experiences. This allows an initial examination of whether the literacy scores correspond to real differences outside the measurement items themselves.

The study nevertheless has several important limitations. First, it uses cross-sectional survey data and therefore cannot establish reliable causal relationships. All demographic, activity and adverse-experience comparisons should be understood as descriptive associations. Second, many variables are self-reported and may be affected by recall error, response bias or differences in respondents’ interpretation of the questions. Third, the main analytical sample for Digital Financial Literacy includes only people with internet access [4]. Conclusions from the Traditional–Digital comparison therefore apply more precisely to internet users than to the entire adult population.

Fourth, the profile analysis depends on the 70% threshold [4]. Converting continuous indicators into “target met” and “target not met” categories helps to interpret and display mismatches, but inevitably discards some continuous information. Two respondents scoring 69% and 70%, for example, are assigned to different categories even though their actual capabilities may differ very little.

Fifth, the study follows the official scoring rules [3] but does not further validate whether the items completely measure the latent concepts of Traditional Financial Literacy and Digital Financial Literacy. Reproducing the official mean scores verifies that the scoring method has been implemented correctly; it does not prove that the indicator itself has unique or optimal conceptual validity.

Finally, the analysis of adverse financial experiences cannot fully separate risk-prevention capability from risk exposure. People who use digital financial services frequently may be more likely to encounter risks even when they also have higher digital financial literacy. This part of the analysis therefore requires particular caution, and the Notebook likewise states explicitly that the comparison should not be given a simple causal interpretation.

7.3 Directions for future research and improvement

Future research could first examine the measurement structure of the financial-literacy indicators in greater depth. The present analysis directly applies the official additive rules for Knowledge, Behaviour and Attitude. Subsequent work could investigate the relationships among these components and between the Traditional and Digital dimensions to determine which specific capabilities account for the observed correlation.

Second, a more systematic robustness analysis could be conducted. Researchers could change the weights assigned to different components, apply other reasonable aggregation methods, or vary the 70% profile threshold and observe whether the main conclusions remain stable. If different indicator constructions produce similar mismatch patterns, confidence in the robustness of the results would increase.

Third, the Traditional–Digital mismatch could be measured more continuously rather than solely through four discrete profiles. For example, researchers could examine which respondents have Digital literacy substantially above or below the level expected from their Traditional literacy. This would identify more precisely those for whom traditional capability has not translated into digital capability.

Fourth, multivariate statistical models could be used to examine demographic characteristics. The current analysis compares age, education, gender and region separately, although these variables may

be correlated with one another. Controlling for multiple factors in the same model would show more clearly whether the observed group differences persist.

Fifth, future analysis of digital activity and adverse experiences could control more fully for the degree of risk exposure. Respondents with similar levels of digital financial use could be compared to determine whether literacy levels still correspond to different adverse-experience rates. This approach would come closer than a direct comparison of the four profiles to addressing whether financial literacy has a protective effect.

Overall, the main contribution of this study is not a single conclusion that financial literacy is “high” or “low”. Instead, it shows that although traditional and digital financial literacy are related, they preserve different information. As financial activity becomes increasingly digital, measuring and reporting the two separately can identify different kinds of capability gaps more effectively than a single composite score and can provide a better foundation for targeted financial education, consumer protection and future research.

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